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SIMATS ENGINEERING

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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ITA0506- COMPUTER VISON

MODEL LAB PRACTICALS

SET 6

1. Perform a 270-degree rotation clockwise along the y-axis for the given image.
2. Perform Affine Transformation on the given image using python and Open CV
3. Perform Perspective Transformation on the given image using python and Open CV.
4. Perform basic Image Handling and processing operations on the image is to read an image in python and detect the corners in the image using Harris Corner Detection function

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cv2. waitKey()
destroy all window ()

Model Lab

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Aim:- To perform a 270-degree rotation clockwise along the Y-axis for given images.

Algorithm:-

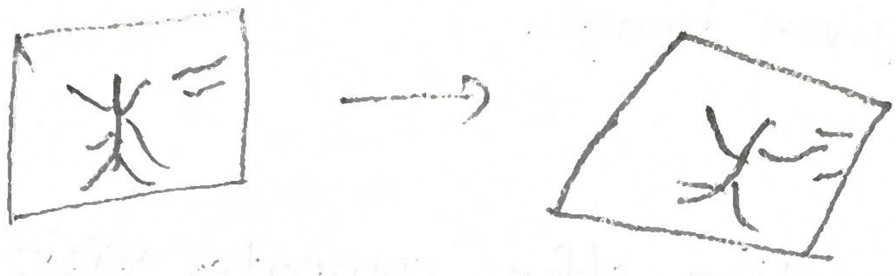
1. Start the importing the computer vision library (cv2)
2. Get the image in the as jpg type or PNG type.
3. Use the function of the rotate of the image.
4. In that give the rotate_270 degrees of the image
5. Finally give the 270 degrees image.

Code:-

```
import cv2
image = cv2.imread("images.jpg")
rimg = cv2.rotated(image, cv2.ROTATE_90_COUTERCLOCKWISE)
cv2.imshow(rimg)
cv2.waitkey(0)
cv2.destroyAll window()
```

cv2. ...
cv2. destroy all window ()

output:-



→ coverting the orginal image into 270deg image.

Result:- The program of 270 degree is rotated the image successfully.

2. Aim:- To perform affine Transformation on the given image using python and OpenCV.

Algorithm:-

→ start importing the computer vision of CV2 libraries.

→ Get the input of the image as a jpg or png type.

→ Use the affine. Transformation function of the

→ Analyze the code properly.

→ Finally get the output.

Code:-

```
import cv2
img = cv2.imread ("image.png")    pt = input(float([[50,50],
                                               [40,400]]))
aimg = cv2.AFFINE TRANSFORM(pt, pts)
wimg = cv2.wrapaffine(aimg, img).
cv2.imshow (wimg)
cv2.waitkey (0)
cv2.destroy Allwindow()
```

Output:-

-) Converting th orginal imag into affine image.

Result:- The program was implemented successfu of the affine transformation.

3. Aim:-

To perform perspective transformation on the given image using python and Open cv,

Algorithm:-

-) start importing the open computer vision library.

-) Get the input of the image as a jpg or png type.

-) Use the transformation function of the perspective.

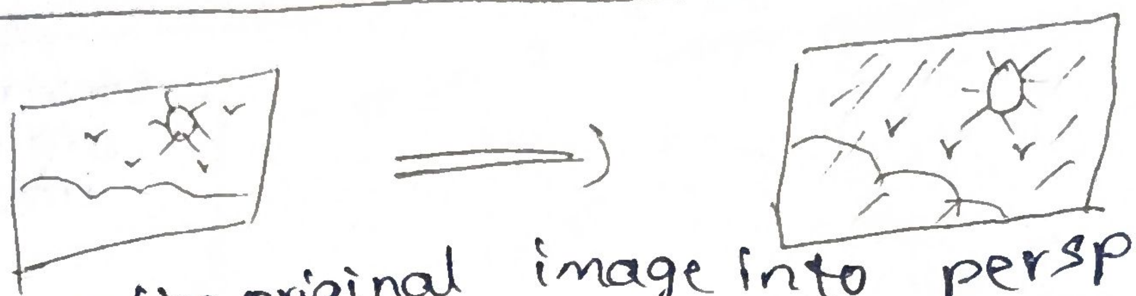
-) Analyze the code properly.

-) finally get the output:

Code:-

```
import cv2
import numpy as ns
img = cv2. imread ("image.jpg")
pt1 = ns. float64 ([[50,50], [60,60]])
pt2 = ns. float64 ([[70,70], [80,80]])
pimg = cv2. get Perspective (pt1, pt2)
wimg = cv2. get Wrap Perspective (img, pimg)
cv2. imshow (wimg)
cv2. waitkey (0)
cv2. destroy all window ()
```

Output:-



Converting original image into perspective image

Result:- The program of perspective transformation of the image was implemented success.

4. Aim:- To perform basic image handling and processing operations on the image is to read an image in python of using Harris Corner Detection.

Algorithm:-

→ Start importing the open computer vision library.

→ Read the input image as a type jpg or png.

→ Use the harris corner function to detect the image.

→ Analyze the code and correct it.

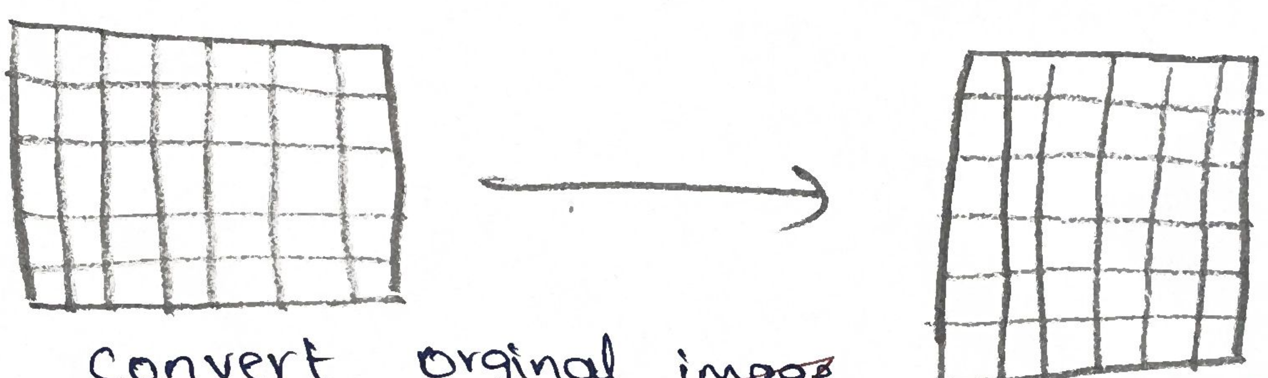
→ Finally get the output image.

Code:-

```
import cv2
img = cv2.imread("image.jpg")
Gray = cv2.cvtCOLOR(img, BGR-2-GRAY)
Himg = cv2.cornorHarris(Gray)
cv2.imshow(Himg)
cv2.waitkey(0)
cv2.destroyAllwindow()
```

blank

output:-



convert orginal image into harris corner image

Result:-

The program uses the harris corner detection was successfully implemented.